

The d- and f-block elements

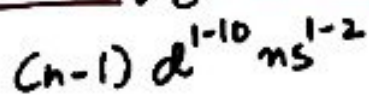
- The d block of the periodic table contain the elements of the group 3-12
- d orbitals are progressively filled in each of the four long period → Transition metal
- The elements constituting the f block are those in which 4f and 5f orbitals are progressively filled in the two long period.
- These elements are formal members of group 3 from which they have been taken out to form a separate f block of periodic table → Inner transition metals

The transition elements (d-block)

Position in the periodic table

- The d block elements lying between s- and p- block → Transition elements
- These elements involve filling of the d orbital of the penultimate energy level in their atoms.
- Four rows of transition elements involve filling of 3d, 4d, 5d and 6d orbital

Electronic configuration



(n-1)d → Inner d orbital
ns → Outermost s orbital

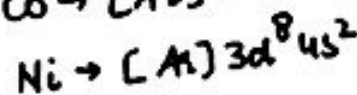
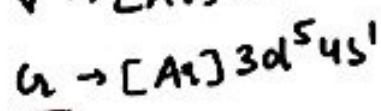
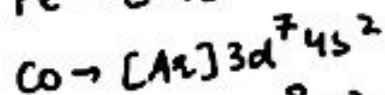
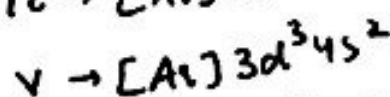
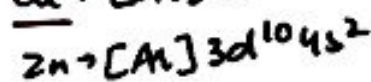
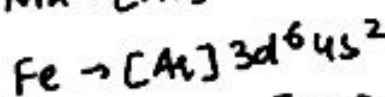
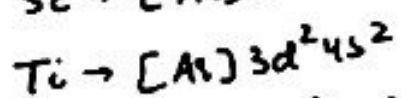
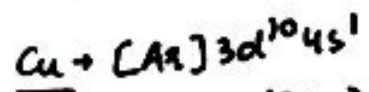
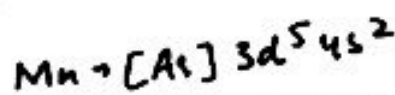
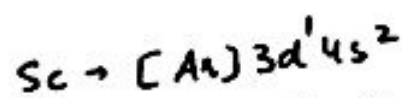
Transition series

1st transition series

→ Involves filling of 3d orbitals

→ Start from Scandium ($Z=21$) and goes upto Zinc ($Z=30$)

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
$Z=21$	22	23	24	25	26	27	28	29	30



→ 4th period → 10 elements

2nd transition series

→ Involve filling of 4d orbital

→ 5th period and 10 elements

→ Yttrium ($Z=39$) to Cadmium ($Z=48$)

Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
$Z=39$	40	41	42	43	44	45	46	47	48

3rd transition series

→ Involve filling of 5d orbitals

→ 6th period and 10 elements

→ First element Lanthanum and 9 elements from Hafnium ($Z=72$) to Mercury ($Z=80$)

La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
$Z=57$	72	73	74	75	76	77	78	79	80

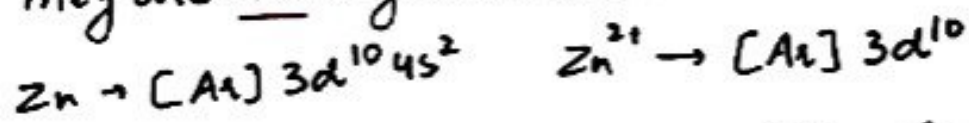
4th transition series

- Filling of 6d orbital
- 7th period and 10 element

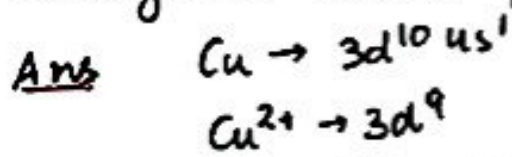
First element Actinium ($Z=89$) and 9 elements from
Rutherfordium to Copernicium
($Z=104$) ($Z=112$)

	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn
$Z =$	89	104	105	106	107	108	109	110	111	112

- Electronic configuration of outer orbitals of Zn, Cd, Hg and Cn are represented by general formula $(n-1)d^{10}ns^2$
- The d orbital in these elements are completely filled in the ground state as well as in their common oxidation state. They are not regarded as transition element



Q Why is copper considered as a transition element even though it has completely filled d orbital?



- Oxidation state has incompletely filled d orbital
→ Transition element

- The d orbitals of the transition elements project to the periphery of an atom more than other orbitals (i.e s and p).
- They are more influenced by the surrounding as well as affecting the atoms or molecules surrounding them.
- Partially filled d orbitals → Exhibit certain characteristic properties →
 - Variable oxidation state
 - Formation of coloured ion
 - Complex formation
 - Catalytic
 - Paramagnetic

General properties of the transition elements

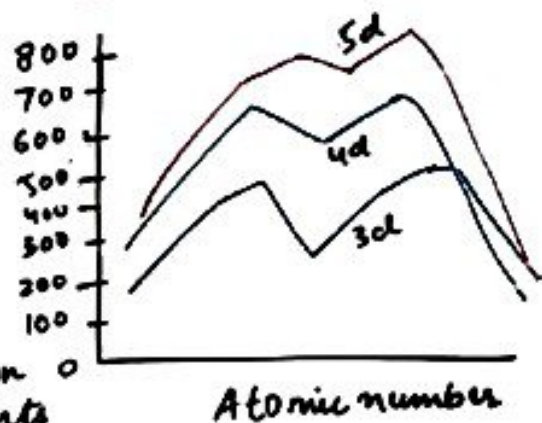
1. Metallic characteristic

- All transition elements are metallic in nature
- They exhibit typical metallic properties such as lustre, high tensile strength, ductility, malleability and high thermal and electrical conductivity.
- Exception of Zn, Cd, Hg and Mn
- Due to presence of strong metallic bond, the transition metals are hard, possess high densities and high enthalpies of atomization
- The enthalpies of atomisation exhibit the maxima at middle of each series

→ 2nd and 3rd series

have greater enthalpies of atomization than 1st series

→ More frequent metal-metal bond in compound of heavy transition elements



Q Give reason ① Zinc has lowest enthalpy of atomisation?

Ans Due to fully filled d orbital (d^{10}). Zn has weak metallic bond, thus has lowest enthalpy of atomisation

Q Why do metals of the second and third series have greater enthalpies of atomisation

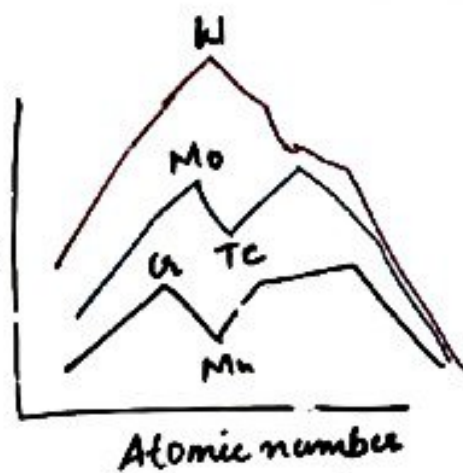
Ans Poor shielding of electrons in 4d and 5d series in comparison to electron of 3d, result in strong metallic bond.

2. Melting and Boiling point

→ d block elements have high Melting and Boiling point

→ Due to presence of unpaired electrons MP/ and partially filled d orbitals 10^{34}

→ They form covalent bond in addition to metallic bond.



→ Manganese (Mn) and Technetium (Tc) → abnormally low melting and boiling point.

→ Chromium (Cr), Molybdenum (Mo), Tungsten (W) high M.P and B.P in their series

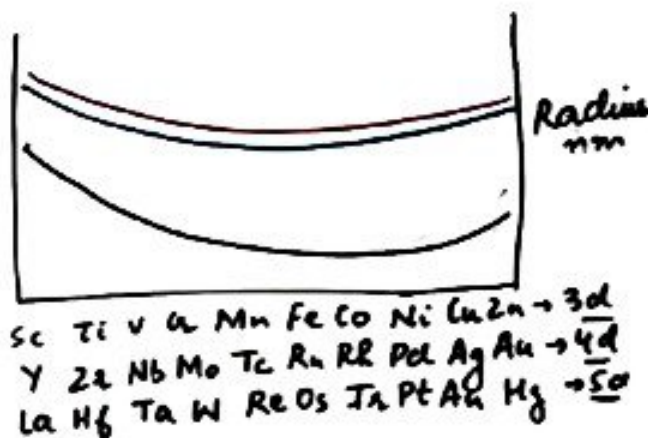
Q Why Zn, Cd and Hg are soft metals?

Ans d orbital completely filled, all electrons are paired. Metallic bonding is weak.

3. Atomic and Ionic radii

→ Among elements of particular transition series, as the atomic number increase, the atomic radii decrease till the middle, becomes almost constant and increase toward end

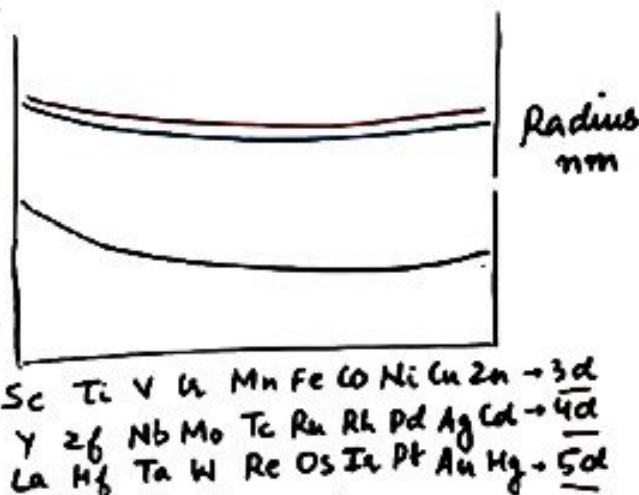
eg 1st series - Radii decrease from Sc to Cr, remain constant till Cu and increase at the end



- Decrease in atomic size at beginning → Increased nuclear charge.
- After middle → Nuclear charge = Screening effect.
- In the end → Repulsion between paired electron dominates
 Nuclear charge < Screening effect
 → Size increases

The atomic radii of 2nd and 3rd transition series are greater as compared to first transition series

→ The elements of 2nd and 3rd transition series belonging to particular group have almost equal atomic radii.

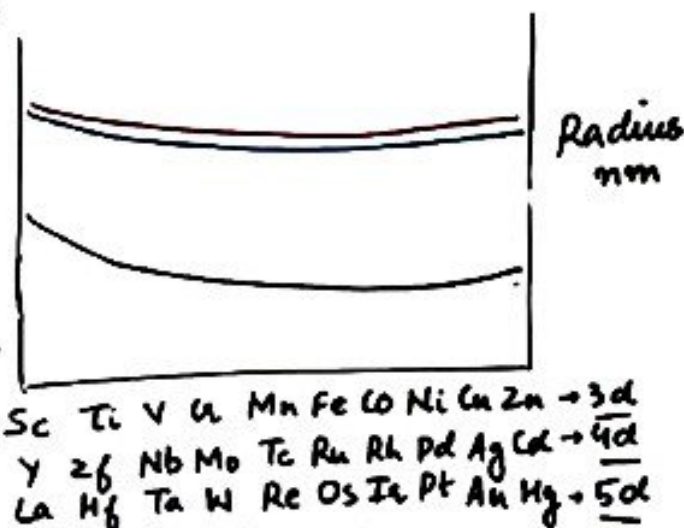


→ This phenomenon is due to the intervention of the 4f orbital which must be filled before the 5d series of elements.

→ Due to addition of new electrons into f subshell and imperfect shielding of one electron by another in f-orbital, there is greater effect of increased nuclear charge than screening effect hence contraction in size occur → Lanthanoid contraction

Q Zirconium and Hafnium are transition series metal of group 4. They are found together in nature and difficult to separate from each other. Give reason.

Ans Zr and Hf have identical radii due to Lanthanoid contraction. That's why they are found together in nature and are difficult to separate.



4. Ionization enthalpy

- In a particular transition series ionization enthalpy increases gradually as we move from left to right.
- The increase in ionization enthalpy is primarily due to increase in nuclear charge.
- As the transition elements involve the gradual filling of (n-1) d orbitals.
- The effect of increase in nuclear charge is partly cancelled by the increase in screening effect.
- Consequently, the increase ionization enthalpy along the period of d-block elements is very small.

- Ionisation enthalpy of 5d series are higher than that of 3d and 4d series due to weak shielding of 4f electron in 5d series.
- Ionization enthalpy depend on
 - Attraction of electron toward nucleus
 - Repulsion between electron
 - Exchange energy.
- Exchange energy → Total number of possible pair of parallel spin in degenerate orbital
 - The loss of energy increases the stability
 - Increase ionization enthalpy



- Ionization enthalpy of Mn^{+} is lower than Cr^{+}

$$Mn^{+} \rightarrow 3d^5 4s^1$$

$$Cr^{+} \rightarrow 3d^5$$

$$Mn^{2+} \rightarrow 3d^5$$

$$Cr^{2+} \rightarrow 3d^4$$
- I.E of Fe^{2+} is lower than Mn^{2+}

$$Fe^{2+} \rightarrow 3d^5 4s^1$$

$$Mn^{2+} \rightarrow 3d^5$$

$$Fe^{3+} \rightarrow 3d^5$$

$$Mn^{3+} \rightarrow 3d^4$$
- Common oxidation state = +2
- 2nd I.E of Cr and Cu high

$$Cr^{+} \rightarrow 3d^5$$

$$Cu^{+} \rightarrow 3d^{10}$$

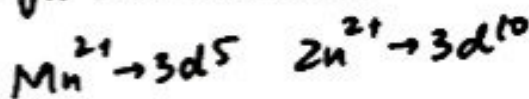
$$Cr^{2+} \rightarrow 3d^4$$

$$Cu^{2+} \rightarrow 3d^9$$
- Zn is corresponding low

$$Zn^{+} \rightarrow 3d^{10} 4s^1$$

$$Zn^{2+} \rightarrow 3d^{10}$$

→ Third ionization enthalpy high for Mn and Zn



Q Give reason Mn^{3+} is a good oxidising agent.

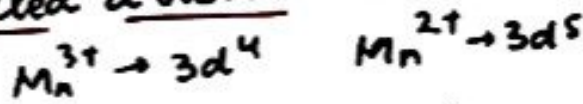
Q Give reason Mn^{3+} is a good oxidising agent because of stability

Ans Mn^{3+} is good oxidising agent because of stability. It will easily

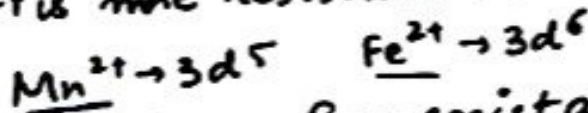
Ans Mn^{3+} is good oxidising agent of Mn^{2+} due to half filled d orbital. It will easily convert to Mn^{2+}

$Mn^{3+} \rightarrow 3d^4$ $Mn^{2+} \rightarrow 3d^5$

$3d^4$ is less stable than $3d^5$ (half filled)



Q. Why is Mn^{2+} more resistant than Fe^{2+} toward oxidation:
 $Mn^{2+} \rightarrow 3d^5$ $Fe^{2+} \rightarrow 3d^6$



Ans Mn^{2+} is stable hence show resistance toward oxidation
 $3d^5$ $3d^5$ is more stable.

→ Fe²⁺ is oxidised easily to form Fe³⁺ which is more stable.

5. Oxidation state

→ All transition metal exhibit a great variety of oxidation state

oxidation state

→ The variable oxidation states of transition elements are due to the participation of ns and (n-1)d electrons in bonding.

Mn	Fe	Co	Ni	Cu	Zn
2	2	3	2	1	0

3d series bonding.

	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
							+2	+2	+1	+2
		+2	+2	+2	+2	+2	+2	+2		
	+3	+3	+3	+3	+3	+3	+3	+3	+2	
		+4	+4	+4	+4	+4	+4	+4		
			+5	+5	+5					
				+6	+6	+6				
					+7					

- In each series, the element which exhibit the maximum number of oxidation state occur either in the middle or near middle of the series eg In 3d series Manganese $+2$ to $+7$.
- The lesser number of oxidation state at the extreme end.
- In beginning of series, they have small number of electrons which they can lose or contribute for sharing eg Sc, Ti
- At end of series, too many d electrons and hence fewer vacant d orbitals can be involved in bonding eg Cu, Zn.
- For the elements of first series (except Scandium) $+2$ oxidation state is the most common oxidation state.
It arises due to loss of 4s electron

→ Oxidation state of transition element differ from each other by unity eg $V \rightarrow +2, +3, +4, +5$ where as oxidation state of non transition element generally differ by two eg $Sn \rightarrow +2, +4$

- P-block elements → lower oxidation state are favoured by heavier members (due to inert pair effect) in case of d block elements higher oxidation state are more stable for heavier member eg group 6 $Mo(VI)$ and $W(VI)$ is more stable than $Cr(VI)$. $Cr(VI)$ in form of dichromate act as strong oxidising agent in acidic medium while MoO_3 and WO_3 are not.

→ Low oxidation state are found when a complex compound has ligands capable of π acceptor character in addition to σ bonding eg Ni(CO)_4 and Fe(CO)_5 , the oxidation state of Ni and Fe is zero

Q In 3d series which element show only +3 oxidation state?
 Ans Sc $\text{Sc}^{3+} \rightarrow 3d^0$ Noble gas configuration

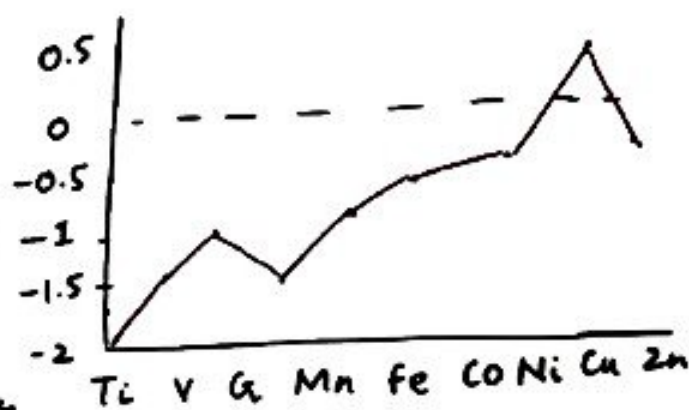
Q Which metal in 3d series exhibit +1 oxidation state?

Ans Cu^+ $\text{Cu}^+ \rightarrow 3d^{10}$
 Stable configuration $\rightarrow$ d orbital fully filled

Trends in the M^{2+}/M Standard electrode potential

→ E° value of M^{2+}/M indicate a decreasing tendency to form divalent cation across the series.

→ Less negative E° value is related to the increase in the sum of first and secondary ionization enthalpy $(I.E_1 + I.E_2)$

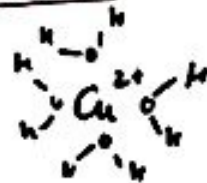


→ Cu having a positive value.

→ The greater energy is required to transform Cu(s) to $\text{Cu}^{2+}(\text{aq})$ which is not balanced by enthalpy of hydration

→ Only oxidizing acids such as Nitric acid and concentrated Sulphuric acid, react with Cu.

In these reactions acid get reduced.



→ E° for Mn, Ni and Zn are more negative than expected from the trend.

→ Stability of half filled d subshell (d^5) in Mn^{2+} .

→ Stability of completely filled d subshell (d^{10}) in Zn^{2+} .

→ In case of Ni, the E° value is related to large negative enthalpy of hydration

Q Why is $E^\circ Mn^{2+}/Mn$ value highly negative as compared to other elements?

Q Why is $E^\circ Cu^{2+}/Cu$ value exceptionally positive?

Ans Ionization enthalpy is very high and hydration enthalpy does not compensate for stabilization of Cu^{2+}

Trends in the M^{3+}/M^{2+} Standard electrode potential

→ Low value of Sc reflect the stability of Sc^{3+}
→ Noble gas configuration

→ Highest value of Zn → Stable d^{10} configuration → Zn^{2+}

→ High value for Mn → Stable d^5 configuration → Mn^{2+}

→ Low value of Fe → Extra stability of Fe^{3+} → d^5

→ Comparatively low of V → Stable V^{2+} → t_{2g}^3 half filled
 $\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array}$ eg

Q Why E° value for Mn^{3+}/Mn^{2+} couple is much more positive than that for Cr^{3+}/Cr^{2+} ?

$\begin{array}{|c|c|c|c|} \hline 7 & 7 & 7 & 7 \\ \hline \end{array}$ t_{2g}

Trends in stability of higher oxidation state

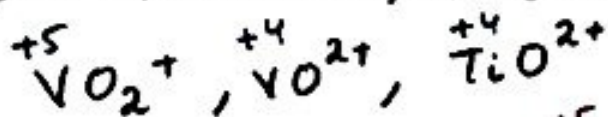
- The highest oxidation state are achieved in TiX_4 ($X = F, Cl, Br, I$), VF_5 and CrF_6 . The +7 oxidation state for manganese is not achieved in simple halide but in MnO_3F .
- Beyond Mn no metal has trihalide except FeX_3 and CoF_3 .
- The ability of fluorine to stabilize the highest oxidation state is due to either higher lattice energy eg CoF_3 or high bond enthalpy for higher covalent compound eg VF_5 , CrF_6 .
- V show oxidation of +5 only in VF_5 . The other pentahalide of V are unstable and undergo hydrolysis to give oxohalides VOX_3 .

- Fluoride are unstable in the low oxidation state eg V exhibit oxidation state of +2 and Cu oxidation state of +1 only in chlorides, bromides and iodides.
- All Cu(II) halides are known except the iodide
 Cu^{2+} oxidises I^- to I_2
$$2Cu^{2+} + 4I^- \rightarrow Cu_2I_2 + I_2$$
- Many Cu(I) compounds are unstable in aqueous solution and undergo disproportionation
$$2Cu^+ \rightarrow Cu^{2+} + Cu$$

 $Cu^{2+}(aq)$ is more stable than $Cu^+(aq)$ due to the much more negative $\Delta_{hyd}H^\circ$ of Cu^{2+} than Cu^+ , which more than compensate for the 2nd ionization enthalpy of Cu.

- The ability of oxygen to stabilize the higher oxidation state is much more than that of fluorine. eg Highest oxidation state of Mn with fluorine is +4 in MnF₄ while +7 in Mn₂O₇
- The superiority of oxygen to fluorine to stabilize the higher oxidation state is due to its ability to form multiple bond with the metal.
- The highest oxidation number in the oxides coincides with group number from Sc₂O₃ to Mn₂O₇
- Beyond group 7 no higher oxides above Fe₂O₃ are known. Fe also form some ferrate (FeO₄)²⁻ in which oxidation state is +6. They are unstable and decompose to give Fe₂O₃ and O₂.

→ Besides the oxides, oxocation stable



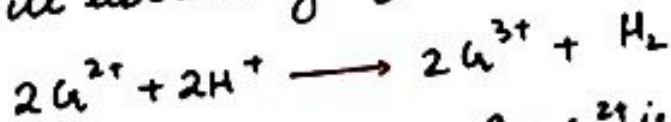
→ Tetrahedral [MO₄]ⁿ⁻ ion $\text{V}^{+5}, \text{Cr}^{+6}, \text{Mn}^{+5}, \text{Mn}^{+6}, \text{Mn}^{+7}$

- Q Why highest fluoride for Mn is MnF₄ whereas highest oxides is Mn₂O₇?
- Q Why Cu⁺ ion is unstable in aqueous solution?
- Q Why highest oxidation state of a metal is shown in its oxides and fluoride?

Chemical reactivity and E° value

- Transition metals differ widely in their chemical reactivity.
- Many metals are sufficiently electropositive to dissolve in mineral acid.
- Some of them remain unaffected by simple acids.
- Metals of first transition series (3d series) with exception of copper have negative E° for M^{2+}/M .
- Metals are more reactive and are oxidised by 1M H^+ .
- However actual rate at which these metal react with oxidizing agent such as H^+ ion is sometime slow.
- Example → Ti and V → Passive to dilute non oxidising agent at room temperature.

- large positive for E° value for redox couple Mn^{3+}/Mn^{2+} , Co^{3+}/Co^{2+} indicate Mn^{3+} and Co^{3+} are strong oxidising agent in aqueous solution.
- The negative value for redox couple Ti^{3+}/Ti^{2+} , V^{3+}/V^{2+} , and Cr^{3+}/Cr^{2+} indicate Ti^{2+} , V^{2+} , Cr^{2+} → strong reducing agent and will liberate hydrogen from dilute acid.



- Q Give reason ① out of d^4 species Cr^{2+} is strong reducing agent while Mn^{3+} is strong oxidising agent

Ans ② Cr^{2+} → Strong reducing agent → Change from d^4 to d^3
half filled t_{2g}^3 configuration.

Mn^{3+} → Strong oxidising agent → Change from d^4 to d^5
→ Half filled d configuration.

Magnetic properties

- The substance which contain some species (atoms, ions or molecules) with unpaired electrons in their orbitals, behave as paramagnetic substance.
- They are weakly attracted by magnetic field.
- The substance whose constituent particle do not contain any unpaired electrons are repelled by magnetic field → Diamagnetic
- The transition metal ions generally contain one or more unpaired electron in them and hence their complex are generally paramagnetic

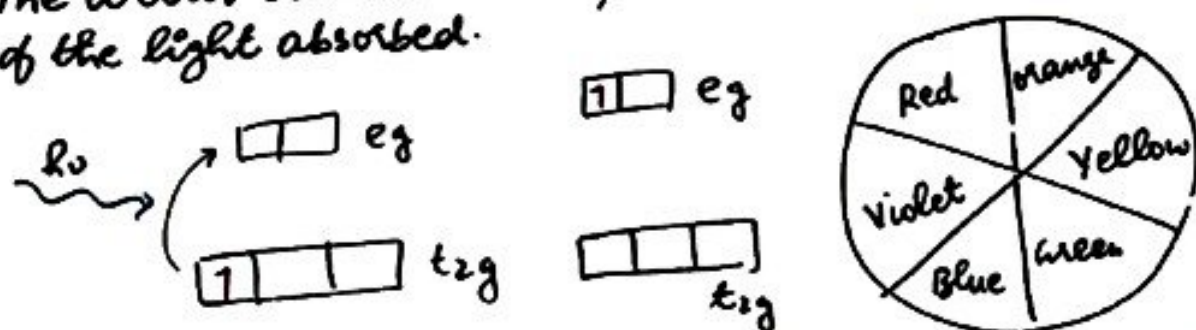
$$\text{Magnetic moment } (\mu) = \sqrt{n(n+2)} \text{ BM}$$

n → Number of unpaired electron
 BM → Bohr magneton.

<u>Metal ion</u>	<u>Configuration</u>	<u>No of unpaired e⁻</u>	<u>Magnetic moment (BM)</u>
Sc^{3+}	$3d^0$	0	
	$3d^1$	1	1.73
Ti^{3+}	$3d^2$	2	2.83
V^{3+}	$3d^3$	3	3.87
Cr^{3+}	$3d^3$	3	4.90
Fe^{2+}	$3d^6$	4	5.92
Mn^{2+}	$3d^5$	5	

Formation of coloured ion

- The compounds of transition elements are usually coloured both in solid state and in aqueous solution
- When an electron from a lower energy d orbital is excited to a higher energy d orbital. → d-d transition
- The energy of excitation corresponds to the frequency of light absorbed.
- The colour observed corresponds to the complementary colour of the light absorbed.



- $Zn^{2+} \rightarrow 3d^{10}$ and $Sc^{3+} \rightarrow 3d^0$ → Salts are white or colourless.
- d-d transition is not possible because in Zn^{2+} all d orbitals are fully filled and in Sc^{3+} all orbitals are vacant.
- Some transition metal in aqueous solution

Metal ion	Configuration	Colour
Sc^{3+}	$3d^0$	Colourless
Ti^{4+}	$3d^0$	Colourless
Ti^{3+}	$3d^1$	Purple
V^{4+}	$3d^1$	Blue
V^{3+}	$3d^2$	Green
Cr^{3+}	$3d^3$	Violet
Mn^{3+}	$3d^4$	

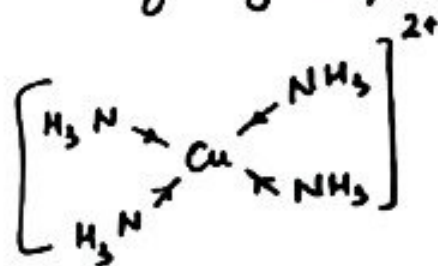
Metal	Configuration	Colour
Mn^{2+}	$3d^5$	Pink
Fe^{3+}	$3d^5$	Yellow
Fe^{2+}	$3d^6$	Green
Co^{2+}	$3d^7$	Pink
Ni^{2+}	$3d^8$	Green
Ni^{3+}	$3d^7$	Blue
Cu^{2+}	$3d^9$	Colourless
Zn^{2+}	$3d^{10}$	

Q Explain ① Cu^{2+} salts are coloured while Zn^{2+} salts are white?

$\underset{3d^9}{Cu^{2+}}$
 $\underset{3d^{10}}{Zn^{2+}}$

Complex formation

- Transition elements form many coordination complex.
- Complex compounds are those in which metal ion bind a number of anion or neutral molecules giving complex species with characteristic properties.
eg $[\text{Fe}(\text{CN})_4]^{2-}$, $[\text{Cu}(\text{NH}_3)_4]^{2+}$
- Their tendency to form complex is attributed to following reason
 - Small size of metal ions
 - High ionic charges
 - Availability of d orbital for bond formation



Catalytic properties

→ Many transition metals and their compounds are known to act as catalyst.

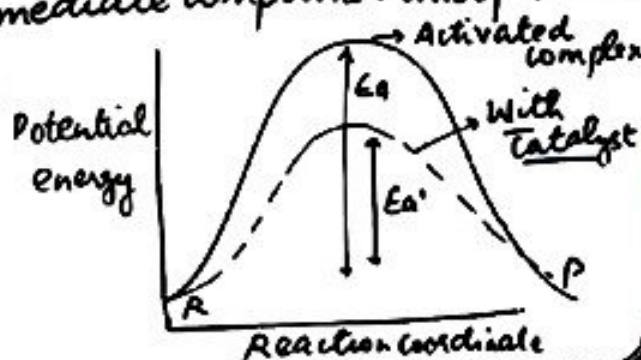
eg Vanadium (V) oxide or V_2O_5 in contact process

finely divided Iron in Haber's process.

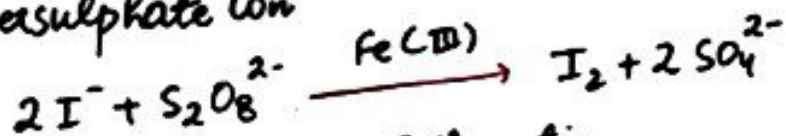
Nickel in catalytic hydrogenation

The catalytic property of transition metal is attributed to the following reason.

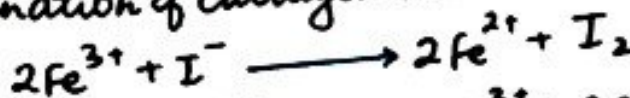
1. Because of their ability to adopt multiple oxidation state and form unstable intermediate compound and provide a new path with lower activation energy for the reaction



eg Iron(III) catalyses the reaction between iodide and persulphate ion



Explanation of catalytic action



2. Transition metal provide a suitable surface for the reaction to take place. The reactant are adsorbed on the surface of the catalyst where reaction occur.

→ Adsorption result in increased concentration of reactant at the surface and also weakens the bond between atoms in reactant molecules.

Formation of Interstitial compounds

→ Interstitial compounds are those which are formed when small atoms like H, C or N trapped inside the crystal lattice of the metal.

→ Usually non stoichiometric and are neither typical ionic or covalent.

Example - TiC , Mn_4N , Fe_3H , $\text{VH}_{0.56}$ and $\text{TiH}_{1.7}$

→ The principal physical and chemical characteristics of these compounds

- (i) They have high melting point, higher than those of pure metal
- (ii) They are very hard, some borides approach diamond in hardness.
- (iii) They retain metallic conductivity
- (iv) They are chemically inert.

Alloy formation

- Alloys are homogeneous solid solutions
- Alloys are formed by atoms with metallic radii that are within about 15 percent of each other
- Transition elements have almost similar atomic size.
- These elements can mutually substitute their position in their crystal lattice.
- Alloys are readily formed by transition metal.
- Hard and high melting point
- Ferrous alloy → Chromium, Vanadium, Tungsten, Molybdenum and manganese used for production of steel and stainless steel.
- Alloy of transition metal with non transition metal
Brass (Copper-Zinc) and Bronze (Copper-Tin)

Some important compounds of transition elements

1. Oxides and oxoanion of metals

- These oxides are formed by the reaction of metal with oxygen at high temperature
- All metal except Scandium form MO oxides which are ionic in nature
- The highest oxidation number of metal in the oxides coincide with the group number from Sc to Mn
(Sc_2O_3) (Mn_2O_7)
- Beyond group 7 no higher oxides except Fe_2O_3
- Oxocation stable → VO_2^+ , VO^{2+} , TiO^{2+}
- These oxides exhibit acidic, basic and amphoteric behaviour.

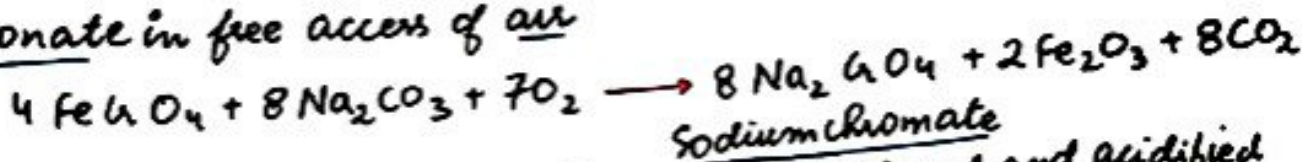
- In general, oxides are basic when the metal is in lower oxidation state and they are acidic if metal is in higher oxidation state
- The oxides are generally amphoteric if the metal is in intermediate oxidation state
- CrO is basic, Cr_2O_3 is amphoteric and CrO_3 is acidic
- V_2O_3 is basic, V_2O_4 (less basic), V_2O_5 (amphoteric).
- V_2O_4 dissolve in acid to give VO^{2+} salt
- V_2O_5 react with alkali to give VO_4^{3-} salt and with acid to give VO_4^{+} salt
- Mn_2O_7 gives HMnO_4 and CrO_3 gives H_2CrO_4 and $\text{H}_2\text{Cr}_2\text{O}_7$ with water

Potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$)

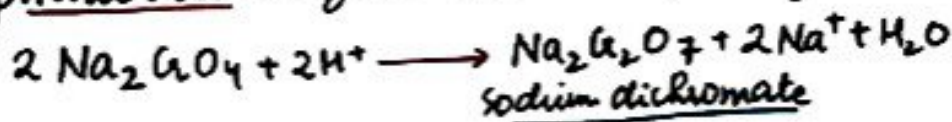
→ Used in leather industry and as an oxidant for preparation of azo compound.

Preparation → They are generally prepared from chromite ore (FeCr_2O_4)

→ Fusion of chromite ore (FeCr_2O_4) with sodium or Potassium carbonate in free access of air



Yellow solution of sodium chromate is filtered and acidified with sulphuric acid to give a solution of orange sodium dichromate



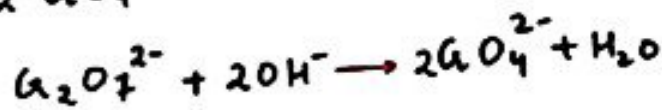
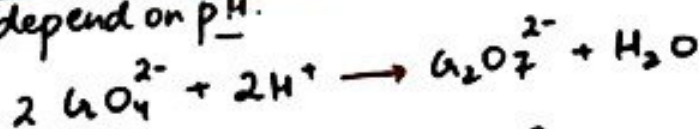
$\text{Na}_2\text{Cr}_2\text{O}_7 \cdot 2\text{H}_2\text{O}$ can be crystallised

- Sodium dichromate is more soluble than potassium dichromate
- The solution containing sodium dichromate is treated with potassium chloride.

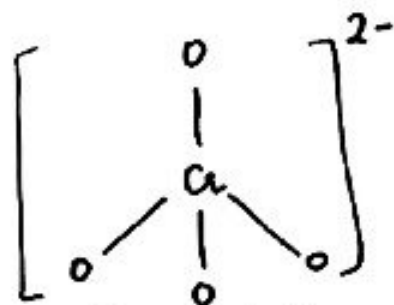


Orange crystal of potassium dichromate crystalline out.

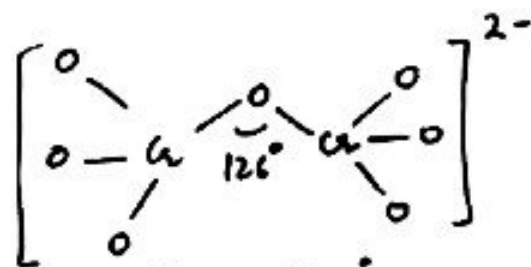
- Chromates and dichromates are interconvertible in aqueous solution depend on pH .



Q Orange colour of $\text{Cr}_2\text{O}_7^{2-}$ ion changes to yellow when treated with alkali Why?



chromate ion
→ Tetrahedral

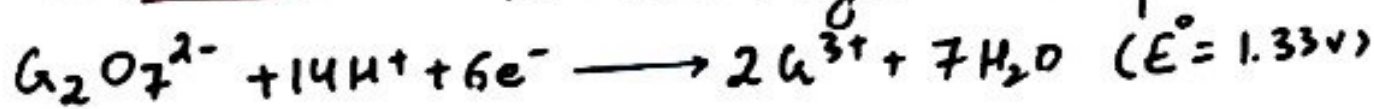


Dichromate ion

- Two tetrahedra
- sharing one corner Cr-O-Cr bond angle 126°

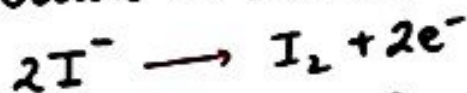
- Sodium and potassium dichromate are strong oxidising agents.
- Sodium salt has greater solubility in water and oxidising agent in organic chemistry
- Potassium dichromate is used as primary standard in volumetric analysis

In acidic solution, its oxidising action is represented as

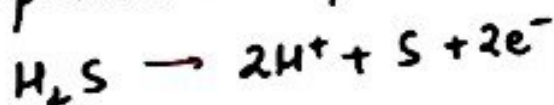


Acidified potassium dichromate will oxidise

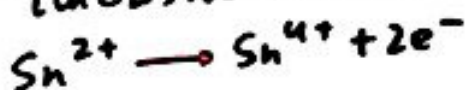
Iodide to Iodine



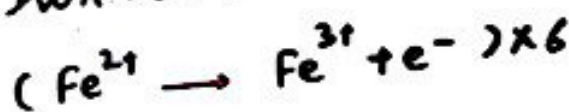
Sulphide to sulphur



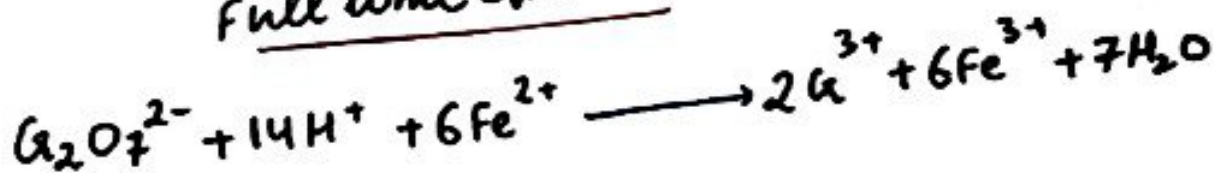
Tin(II) to Tin(IV)



Iron(II) to Iron(III)



Full ionic equation



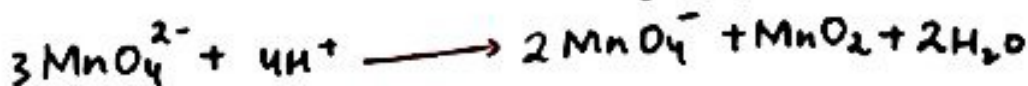
Potassium permanganate (KMnO_4)

→ It is prepared by fusion of MnO_2 with alkali metal hydroxide in presence of atmospheric oxygen or an oxidizing agent like potassium nitrate to produce K_2MnO_4

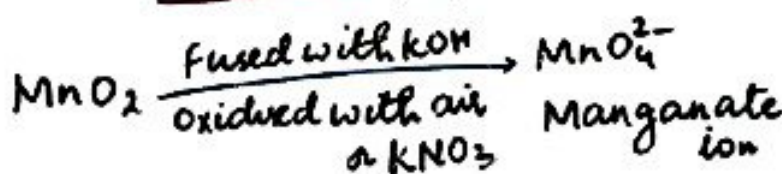


Potassium manganate (green)

→ K_2MnO_4 is disproportionate in acidic or neutral solution to give permanganate



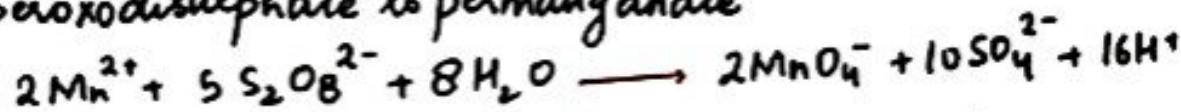
Commercial preparation



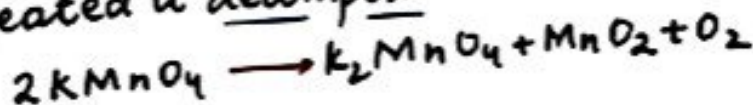
Manganate ion

Permanganate ion

In a laboratory, a manganese (II) ion salt is oxidized by peroxodisulphate to permanganate

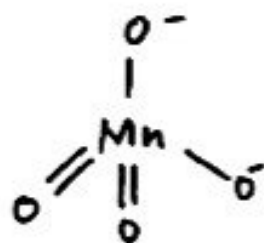
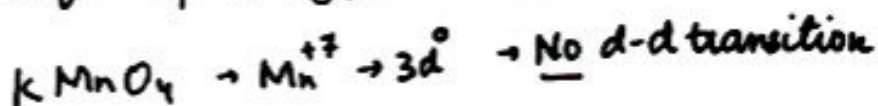


- Potassium permanganate form dark purple crystal
- Is structural with KClO₃
- Salt is not very soluble in water (6.4 g/100g of water at 293K)
- But when heated it decompose at 513K

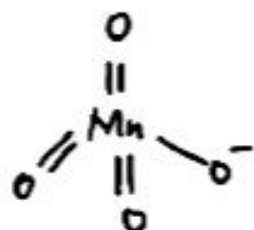


- color of KMnO_4 is due ligand to metal (L → M)

charge transfer 2p of oxygen to 3d of Manganese



Manganate ion (Green)



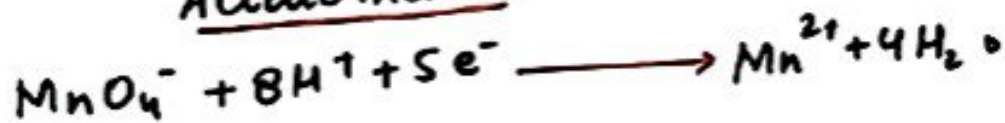
Permanganate ion (Purple)

- Both are tetrahedral
- Manganate ion → Paramagnetic with one unpaired electron
- Permanganate ion → Diamagnetic

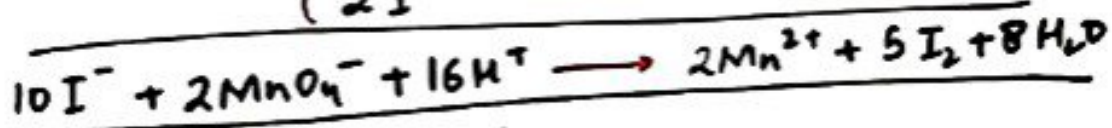
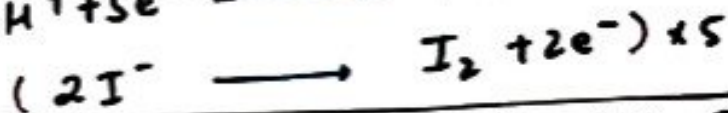
π bonding take place by overlap of p orbital of oxygen with d orbital of manganese.

KMnO_4 act as strong oxidizing agent.

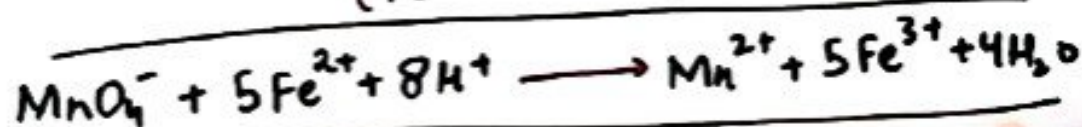
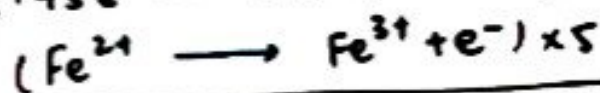
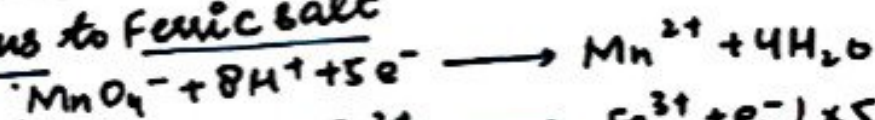
Acidic medium



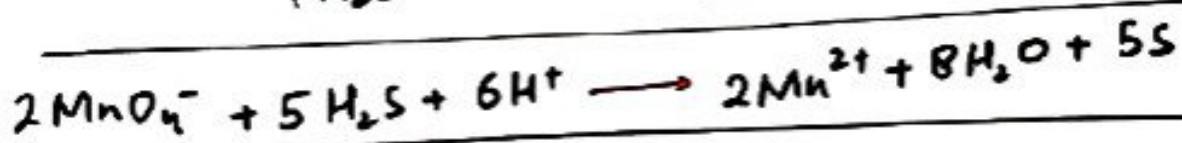
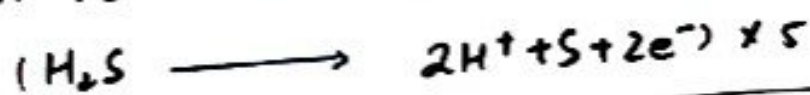
① Iodide to Iodine



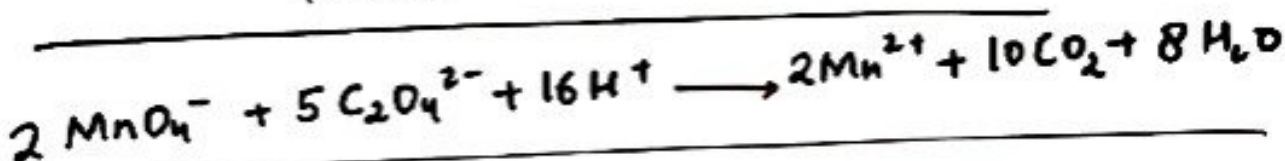
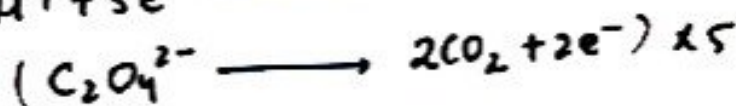
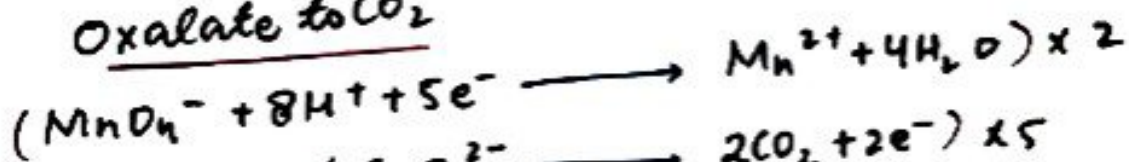
② Ferrous to Ferric salt



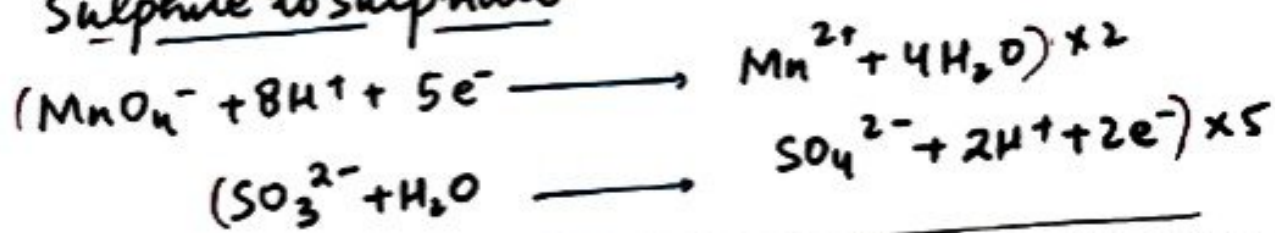
③ H_2S to Sulphur



④ Oxalate to CO_2



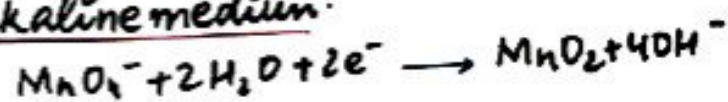
⑤ Sulphite to sulphate



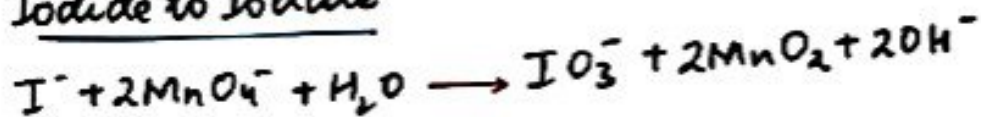
⑥ Nitrite to Nitrate



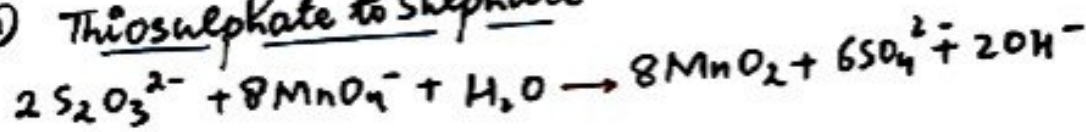
2. In neutral or faintly alkaline medium.



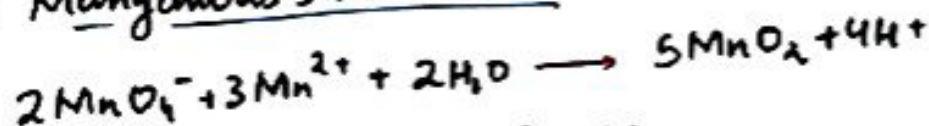
① Iodide to Iodate



② Thiosulphate to Sulphate

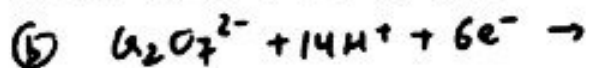
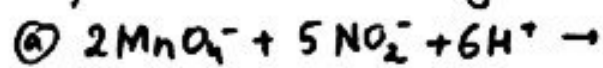


③ Manganous salt to MnO₂

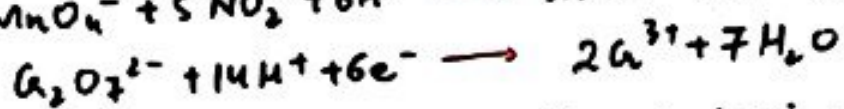
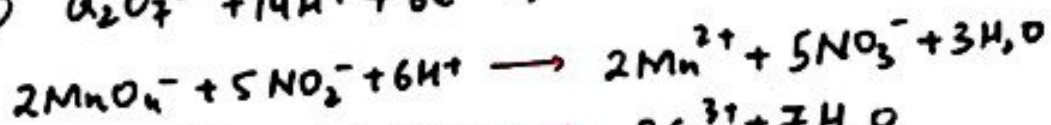


- Uses
- used in analytical chemistry
 - used as oxidant in preparation of organic compounds
 - Bleaching of wool, cotton and silk
 - Decolourisation of oil

Q Complete the following reaction

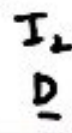
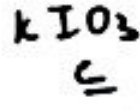
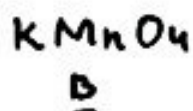
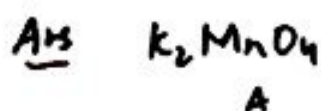


Ans



Q Explain the method of preparation of potassium dichromate from chromite ore?

Q When MnO_2 is fused with KOH in presence of KNO_3 as oxidising agent. It gives a dark green compound A. Compound A disproportionates in acid to give compound B. An alkaline solution of B oxidise KI to C and acidified solution of C oxidises KI to D. Find A, B, C and D?



The inner transition element (f block)

→ Elements in which the last electron fill up in f orbital are called f-block elements

They have two series

- Lanthanoids (14 elements following Lanthanum)
 - Lanthanum itself does not possess any 4f electron
 - Include in discussion of Lanthanoid series
- Actinoids (14 elements following Actinium)
 - Actinium is studied along with actinoids

Lanthanoids.

Electronic configuration $\rightarrow 4f^{1-14} 5d^{0-1} 6s^2$

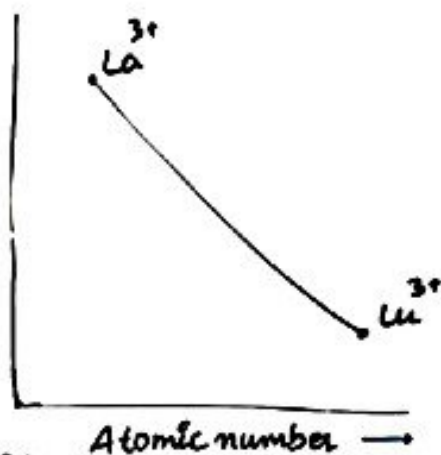
57 Lanthanum La	$5d^1 6s^2$	64 Gadolinium Gd	$4f^7 5d^1 6s^2$
58 Cerium Ce	$4f^1 5d^1 6s^2$	65 Terbium Tb	$4f^9 6s^2$
59 Praseodymium Pr	$4f^3 6s^2$	66 Dysprosium Dy	$4f^{10} 6s^2$
60 Neodymium Nd	$4f^4 6s^2$	67 Holmium Ho	$4f^{11} 6s^2$
61 Promethium Pm	$4f^5 6s^2$	68 Erbium Er	$4f^{12} 6s^2$
62 Samarium Sm	$4f^6 6s^2$	69 Thulium Tm	$4f^{13} 6s^2$
63 Europium Eu	$4f^7 6s^2$	70 Ytterbium Yb	$4f^{14} 6s^2$
		71 Lutetium Lu	$4f^{14} 5d^1 6s^2$

Cerium, Gadolinium and Lutetium have electron in 5d.

Atomic and Ionic size

$\rightarrow$ Atomic radii and Ionic radii of Lanthanoid (M^{3+}) decreases steadily from lanthanum to lutetium

↑
Ionic
radii
/pm



$\rightarrow$ The decrease in atomic radii is not quite regular as it is regular in M^{3+} ions

$\rightarrow$ As the atomic number increases in Lanthanoid series, electron goes to $4f$ orbital

$\rightarrow$ The $4f$ electrons constitute inner shell and are ineffective in screening the nuclear charge.

$\rightarrow$ Shielding of one $4f$ electron is less than one d electron

$\rightarrow$ Increase in nuclear charge along the series and decrease in size $\rightarrow$ Lanthanoid contraction

Consequence of Lanthanoid contraction

① Similarity in size of elements of 2nd and 3rd transition series

The usual increase in size on moving down the group from 2nd to 3rd transition element is cancelled by the decrease in size due to lanthanoid contraction eg Zr (160 pm) and Hf (159 pm) have nearly equal atomic radii.

② Separation of Lanthanoid

→ Due to lanthanoid contraction there is small difference in size of lanthanoids

→ They have quite similar properties and due to this reason they are difficult to separate.

③ Basic character of oxides and hydroxide

Due to lanthanoid contraction, the covalent nature of $Ln-OH$ bond increases as the basic character of oxides and hydroxides decrease from $La(OH)_3$ to $Lu(OH)_3$

Oxidation state

- Most stable oxidation state of lanthanoid is +3.
- +2 and +4 also obtained in solution or solid compound.
- Gadolinium and lutetium are especially stable in +3
half filled completely filled 4f subshell.
- Cerium and Terbium exhibit oxidation state of +4.
- $Ce^{4+} \rightarrow 4f^0$ $Tb^{4+} \rightarrow 4f^7$
- $Eu^{2+} \rightarrow 4f^7$ $Yb^{2+} \rightarrow 4f^{14}$ also exist in aqueous solution
- Samarium (Sm) and Euprium (Eu) $\rightarrow$ +2 +3 oxidation state
- Few lanthanoid exhibit +4 and +2 oxidation state yet they have tendency to attain +3 oxidation state $\rightarrow$ stable
- $Ce^{4+} \rightarrow$ Good oxidising agent $Sm^{2+} \rightarrow$ Good reducing agent

General characteristics

1. Appearance $\rightarrow$ lanthanoid are silvery white soft metal
 - $\rightarrow$ Tarnish rapidly in air
 - $\rightarrow$ Hardness increase with increasing atomic number.
 - $\rightarrow$ Samarium being steel hard.
2. Density and melting point
 - $\rightarrow$ Dense metal and have high melting point (1000 to 1200 K)
 - $\rightarrow$ Sm has exceptionally high melting point $\rightarrow$ 1623 K
3. Coloured ions $\rightarrow$ Coloured in solid as well as in solution
 - $\rightarrow$ Partially filled f orbital $\rightarrow$ f-f transition
 - $\rightarrow$ Absorption bands are narrow
 - $\rightarrow$ $La^{3+} (4f^0)$ and $Lu^{3+} (4f^{14}) \rightarrow$ Do not show any colour.

4. Magnetic behaviour → Paramagnetic

→ Due to unpaired electrons in f orbitals

→ $4f^0$ (La^{3+} , Ce^{4+}) and $4f^{14}$ (Yb^{2+} , Lu^{3+}) → Not paramagnetic

→ Maximum in Neodymium.

5. Ionization enthalpy

→ First ionization enthalpy → 600 kJ mol^{-1}

→ Second ionization enthalpy → 1200 kJ mol^{-1}

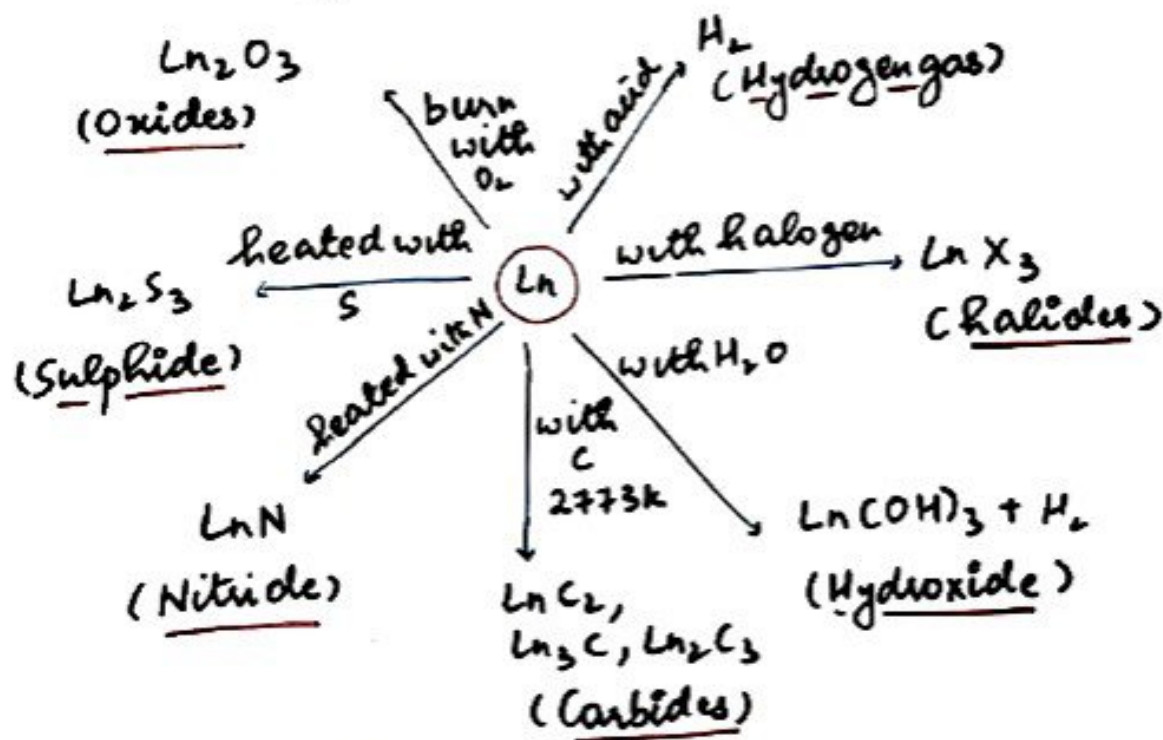
Comparable with Ca

→ Third ionization enthalpy → Variation → Stability

low value → $\frac{\text{Lanthanum}}{4f^0}$, $\frac{\text{Gadolinium}}{4f^7}$, $\frac{\text{Lutetium}}{4f^{14}}$

Chemical behaviour

→ Earlier member of series are quite reactive similar to Ca but increasing atomic number they behave more like Al.



Uses.

1. Lanthanoids are used for the production of alloy steel for plates and pipes.
2. Mischmetall $\rightarrow$ alloy of a lanthanoid metal ($\sim 95\%$) and Iron ($\sim 5\%$) and traces of S, C, Ca and Al.
 $\rightarrow$ Mischmetall is used in Mg based alloy to produce bullet, shell and lighter flint.
3. Mixed oxides of lanthanoid are used as catalyst in cracking of petroleum.
4. Some oxides are used in phosphors in television screen and fluorescing surface.

Q Write two consequence of lanthanoid contraction?

Ans ① Similarity in atomic size of 2nd and 3rd transition series in same group eg Zr and Hf.

② There is small difference in size of lanthanoids. They have similar properties and difficult to separate.

Q Why Ce^{4+} is good oxidising agent while Sm^{2+} is a good reducing agent?

Ans Lanthanoids are most stable in +3 oxidation state

$\rightarrow \text{Ce}^{4+} \rightarrow \text{Ce}^{3+} \rightarrow$ Good oxidising agent
 $\rightarrow \text{Sm}^{2+} \rightarrow \text{Sm}^{3+} \rightarrow$ Good reducing agent

Q Electronic configuration of Ce = $[\text{Xe}] 4f^1 5d^1 6s^2$
Spin only magnetic moment of Ce^{3+} ?

Ans $\text{Ce}^{3+} \rightarrow [\text{Xe}] 4f^1$ $\mu = \sqrt{n(n+2)} = \sqrt{1(1+2)} = \sqrt{3} = 1.73 \text{ BM}$

The Actinoids.

- The series of elements involve the filling of 5f orbital is called actinoid series.
- It include 14 elements from Thorium (Th) to Lawrencium (Lr)
- Radioactive elements
- Earlier member have relatively long half lives.
- The latter one have half life value ranging from a day to 3 minutes for Lawrencium (Lr)
- They are prepared in nanogram quantities
- Study is difficult

Electronic configuration

$5f^{1-14} 6d^{0-1} 7s^2$

89 Actinium	Ac	$6d^1 7s^2$	96 Curium	Cm	$5f^7 6d^1 7s^2$
90 Thorium	Th	$6d^2 7s^2$	97 Berkelium	Bk	$5f^9 7s^2$
91 Protactinium	Pa	$5f^2 6d^1 7s^2$	98 Californium	Cf	$5f^{10} 7s^2$
92 Uranium	U	$5f^3 6d^1 7s^2$	99 Einsteinium	Es	$5f^{11} 7s^2$
93 Neptunium	Np	$5f^4 6d^1 7s^2$	100 Fermium	Fm	$5f^{12} 7s^2$
94 Plutonium	Pu	$5f^6 7s^2$	101 Mendelevium	Md	$5f^{13} 7s^2$
95 Americium	Am	$5f^7 7s^2$	102 Nobelium	No	$5f^{14} 7s^2$
			103 Lawrencium	Lr	$5f^{14} 6d^1 7s^2$

Irregularities in electronic configuration due to stability of $5f^7$, $5f^{14}$.

Ionic size → There is gradual decrease in the size of atom or M^{3+} ion across the series

→ The steady decrease in ionic radii with increase in atomic number → Actinoid contraction

→ The contraction is greater from element to element in this series. This is due to poor shielding by 5f electrons

Oxidation state

→ Greater range of oxidation state

→ 5f, 6d, 7s level are of comparable energies

→ Most common oxidation state +3

→ Maximum oxidation at Th (+4), Pa (+5),
U (+6), Np (+7)

Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md
3		3	3	3	3	3	3	3	3	3	3	3
	4	4	4	4	4	4	4	4				
		5	5	5	5	5						
			6	6	6	6						
				7	7							

→ Actinoid resemble the lanthanoid in having more common +3 oxidation than in the +4.

→ However +3 and +4 tend to hydrolyse.

Similarities between Lanthanoid and Actinoids

- Both exhibit +3 oxidation state predominantly.
- Both are electropositive and high reactivity.
- like lanthanoid contraction, there is actinoid contraction.
- In both series f orbitals are being progressively filled.

Difference

Lanthanoids

1. Beside common oxidation state of +3, some lanthanoid exhibit common oxidation state of +2 and +4.
2. They have smaller tendency to form complex.
3. They do not form oxo ion.
4. Except Promethium they are non radioactive.
5. Compounds are less basic.

Actinoids

1. Besides common oxidation state of +3. They show higher oxidation state of +4, +5, +6 and +7.
2. They have strong tendency to form complex.
3. They do form oxo ions
 UO_2^+ , PuO_2^+ , VO^+
4. All the actinoid are radioactive.
5. Compounds are more basic.

Q Why do actinoid show a wide range of oxidation state.

Ans Actinoid have a greater range of oxidation state due to comparable energies of 5f, 6d and 7s orbitals. Hence all their electrons can take part in bond formation.

Q Why actinoid contraction is greater than Lanthanoid contraction?

Ans 5f electrons are more poorly shielded than 4f electrons.

Q Chemistry of actinoids is complicated as compared to Lanthanoids. Give two reasons.

Ans (i) Actinoid display large number of oxidation state
(ii) Actinoids are radioactive in nature.

Q Give three point of difference between Lanthanoids and actinoids.